

# The ideal interpolation pulse

## Background

The normalized sinc is  $\text{sinc}(u) = \sin(\pi u)/(\pi u)$  for  $u \neq 0$ , and  $\text{sinc}(0) = 1$ . For  $T=1/f_s$ , the ideal interpolation pulse is  $h(t)=\text{sinc}(t/T)$ . It is one at  $t=0$  and zero at all other integer multiples of  $T$ .

## Your task

Evaluate the normalized sinc, including its value at zero. Then evaluate  $h(t)$  for a given  $f_s$ . Display the supplied plot and print the pulse values at  $t = [-2T, -T, 0, T, 2T]$ .

## Function contract and assumptions

`normalized_sinc(u)` returns an array matching the shape of the real input array  $u$ . `ideal_pulse(t, fs)` returns  $h(t)$ , with the same shape as  $t$ . Assume  $f_s > 0$ . For this problem only, do not use a built-in sinc function.

## Example / correctness check

`normalized_sinc([0, 0.5, 1, -1])` is approximately `[1, 0.63661977, 0, 0]`. For every valid  $f_s$ , `h([-2T,-T,0,T,2T])` is approximately `[0,0,1,0,0]`.

## Exam instructions

Complete only the TODO functions in `template.py` and preserve their signatures. The supplied driver handles printing and plotting. Your functions must work for all valid inputs described above, not just the example. No external files or packages beyond NumPy and Matplotlib are needed. Small floating-point differences are acceptable.

Topic reference: PDF pages 25-27.